Oliver H. E. Philcox, Ph.D.

Physics & Astrophysics Building, 452 Lomita Mall, Stanford, CA 94305, USA

Website: oliverphilcox.github.io

Email: ohep2@cantab.ac.uk, ophilcox@stanford.edu

CURRENT POSITION

Department of Physics, Stanford University, Stanford, USA *2025 - Present*
Assistant Professor

Leinweber Institute for Theoretical Physics at Stanford, *Faculty*
Kavli Institute for Particle Astrophysics and Cosmology, *Senior Member*

PREVIOUS POSITIONS

Simons Society of Fellows, New York, USA *2022 - 2025*
Junior Fellow

Department of Physics, Columbia University, New York, USA *2022 - 2025*
Postdoctoral Research Scientist

Mentors: Prof. J. Colin Hill & Prof. Lam Hui

EDUCATION

Department of Astrophysical Sciences, Princeton University, USA *2019 - 2022*
Ph.D. in Astrophysics (2022)

Thesis: 'Probing Fundamental Cosmology with Galaxy Surveys'

Thesis Advisors: David N. Spergel & Matias Zaldarriaga

M.A. in Astrophysics (2020)

Center for Astrophysics | Harvard & Smithsonian, Cambridge, USA *2018 - 2019*
Pre-Doctoral Student, *Herchel-Smith Scholar*

Advisor: Daniel J. Eisenstein

Institute of Astronomy, University of Cambridge, UK *2017 - 2018*
M.Sci. in Astrophysics

Part III: 1st Class (Rank 1/28, 97%)

Thesis: 'Detection and Removal of B-mode CMB Dust Foregrounds with Signatures of Statistical Anisotropy'

Thesis Advisor: Blake D. Sherwin

Emmanuel College, University of Cambridge, UK *2014 - 2017*
M.A. (Cantab.) in Natural Sciences, *Senior Scholar*

Parts IA, IB, II: 1st Class (Rank 1/20, 90%)

LONG-TERM ACADEMIC VISITS

Center for Computational Astrophysics Jul. 2021 - Aug. 2025
Guest Researcher *New York, USA*

Institute for Advanced Study Sep. 2020 - Jul. 2022
Visiting Graduate Student with Matias Zaldarriaga *Princeton, USA*

Max-Planck Institute for Astrophysics Aug. - Sep. 2020
Visiting Graduate Student with Eiichiro Komatsu *Munich, Germany*

Department of Applied Mathematics and Theoretical Physics May - Jul. 2020
Visiting Graduate Student with Blake D. Sherwin *Cambridge, UK*

Max-Planck-Institut für Astronomie Jul. - Sep. 2017
Summer Intern with Jan Rybizki *Heidelberg, Germany*

Center for Astrophysics | Harvard & Smithsonian Jun. - Aug. 2016
Undergraduate Research Fellow with Ákos Bogdán *Cambridge, USA*

PUBLICATION LIST

* = *Author list alphabetized*

Major Author

1. Ivanov, M. M., Sullivan, J. M., Chen, S.-F., Chudaykin, A., Maus, M., **Philcox, O. H. E.**, "Reanalyzing DESI DR1: 4. Percent-Level Cosmological Constraints from Combined Probes and Robust Evidence for the Normal Neutrino Mass Hierarchy", *submitted to Phys. Rev. D* (arXiv).
2. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, "Reanalyzing DESI DR1: 3. Constraints on Inflation from Galaxy Power Spectra & Bispectra", *submitted to Phys. Rev. D* (arXiv).
3. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, "Reanalyzing DESI DR1: 2. Constraints on Dark Energy, Spatial Curvature, and Neutrino Masses", *submitted to Phys. Rev. D* (arXiv).
4. **Philcox, O. H. E.**, Zhong, K., Sirletti, S. S., "Separating the Inseparable: Constraining Arbitrary Primordial Bispectra with Cosmic Microwave Background Data", *submitted to Phys. Rev. D* (arXiv).
5. El-Haj, L. H. A., **Philcox, O. H. E.**, Hill, J. C., "Constraining Inflationary Particle Production with CMB Polarization", *submitted to Phys. Rev. D* (arXiv).
6. *Bakx, T., Ivanov, M. M., **Philcox, O. H. E.**, Vlah, Z., "One-Loop Galaxy Bispectrum: Consistent Theory, Efficient Analysis with COBRA, and Implications for Cosmological Parameters", *submitted to Phys. Rev. D* (arXiv).
7. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, "Reanalyzing DESI DR1: 1. ΛCDM Constraints from the Power Spectrum and Bispectrum", *accepted by Phys. Rev. D* (arXiv).
8. ***Philcox, O. H. E.**, Silverstein, E., Torroba, G., "Quantum stress-energy at timelike boundaries: testing a new beyond-ΛCDM parameter with cosmological data", *accepted by JCAP* (arXiv).
9. **Philcox, O. H. E.**, Hill, J. C., "Integrated Sachs-Wolfe lensing bispectrum and trispectrum", *Phys. Rev. D* **111**, 123554 (2025) (arXiv).
10. Goldstein, S., **Philcox, O. H. E.**, Fondi, E., Coulton, W. R., "Wonderings on wiggly bispectra: Nonlinear evolution and reconstruction of oscillations in the squeezed bispectrum", *Phys. Rev. D* **112**, 083503 (2025) (arXiv).
11. Zhang, G., Modi, C., **Philcox, O. H. E.**, "Modeling galaxy surveys with hybrid simulation-based inference", *Phys. Rev. D* **113**, 043518 (2026) (arXiv).
12. **Philcox, O. H. E.**, "Searching for inflationary physics with the CMB trispectrum. III. Constraints from *Planck*", *Phys. Rev. D* **111**, 123534 (2025) (arXiv).
13. **Philcox, O. H. E.**, "Searching for inflationary physics with the CMB trispectrum. II. Code and validation", *Phys. Rev. D* **111**, 123533 (2025) (arXiv).
14. **Philcox, O. H. E.**, "Searching for inflationary physics with the CMB trispectrum. I. Primordial theory and optimal estimators", *Phys. Rev. D* **111**, 123532 (2025) (arXiv).
15. ***Philcox, O. H. E.**, Shiraishi, M., "Non-Gaussianity beyond the scalar sector: A search for tensor and mixed tensor-scalar bispectra with Planck data", *Phys. Rev. D* **111**, 123502 (2025) (arXiv).
16. Goldstein, S., **Philcox, O. H. E.**, Hill, J. C., Hui, L., "Intermediate mass-range particles from small scales: Nonperturbative techniques for cosmological collider physics from large-scale structure surveys", *Phys. Rev. D* **110**, 083516 (2024) (arXiv).

17. *Coulton, W. R., **Philcox, O. H. E.**, Villaescusa-Navarro, F., "The Impact of Non-Gaussian Primordial Tails on Cosmological Observables", *submitted to Phys. Rev. D* (arXiv).

18. *Chen, S.-F., Ivanov, M. M., **Philcox, O. H. E.**, Wenzl, L., "Suppression without Thawing: Constraining Structure Formation and Dark Energy with Galaxy Clustering", *Phys. Rev. Lett.* **133**, 231001 (2024) (arXiv).

19. **Philcox, O. H. E.**, Kumar, S., Hill, J. C., "Searching for inflationary particle production in Planck data", *Phys. Rev. D* **111**, 103523 (2025) (arXiv).

20. **Philcox, O. H. E.**, Flöss, T., "Suite of optimal and efficient power spectrum and bispectrum estimators for large-scale structure analyses", *Phys. Rev. D* **112**, 063507 (2025) (arXiv).

21. Cabass, G., **Philcox, O. H. E.**, Ivanov, M. M., Akitsu, K., Chen, S.-F., Simonović, M., Zaldarriaga, M., "BOSS constraints on massive particles during inflation: The cosmological collider in action", *Phys. Rev. D* **111**, 063510 (2025) (arXiv).

22. de Belsunce, R., **Philcox, O. H. E.**, Irsic, V., McDonald, P., Guy, J., Palanque-Delabrouille, N., "The 3D Lyman-α forest power spectrum from eBOSS DR16", *MNRAS* **533**, 3756 – 3770 (2024) (arXiv).

23. **Philcox, O. H. E.**, Ereza, J., "Could Sample Variance be Responsible for the Parity-Violating Signal Seen in the BOSS Galaxy Survey?", *Phil. Trans. Roy. Soc. A* **383**, 2290 (2025) (arXiv).

24. ***Philcox, O. H. E.**, Shiraishi, M., "Testing graviton parity and Gaussianity with Planck T-, E-, and B-mode bispectra", *Phys. Rev. D* **109**, 063522 (2024) (arXiv).

25. Goldstein, S., **Philcox, O. H. E.**, Hill, J. C., Esposito, A., Hui, L., "Consistently constraining $f_{\rm NL}$ with the squeezed lensing bispectrum using consistency relations", *Phys. Rev. D* **109**, 043515 (2024) (arXiv).

26. *Modi, C., **Philcox, O. H. E.**, "Hybrid SBI or How I Learned to Stop Worrying and Learn the Likelihood", *submitted to Phys. Rev. Lett.* (arXiv).

27. **Philcox, O. H. E.**, König, M. J., Alexander, S., Spergel, D. N., "What can galaxy shapes tell us about physics beyond the standard model?", *Phys. Rev. D* **109**, 063541 (2024) (arXiv).

28. ***Philcox, O. H. E.**, Shiraishi, M., "Testing parity symmetry with the polarized cosmic microwave background", *Phys. Rev. D* **109**, 083514 (2024) (arXiv).

29. *Coulton, W. R., **Philcox, O. H. E.**, Villaescusa-Navarro, F., "Signatures of a parity-violating universe", *Phys. Rev. D* **109**, 023531 (2024) (arXiv).

30. **Philcox, O. H. E.**, "Optimal estimation of the binned mask-free power spectrum, bispectrum, and trispectrum on the full sky: Tensor edition", *Phys. Rev. D* **108**, 063506 (2023) (arXiv).

31. *Ivanov, M. M., **Philcox, O. H. E.**, "Measuring H_0 with Spectroscopic Surveys", *chapter in "Hubble Constant Tension" (Eds. Di Valentino, E. and Brout, D., Springer, Singapore)* (arXiv).

32. **Philcox, O. H. E.**, "Do the CMB Temperature Fluctuations Conserve Parity?", *Phys. Rev. Lett.* **131**, 181001 (2023) (arXiv).

33. **Philcox, O. H. E.**, "Optimal estimation of the binned mask-free power spectrum, bispectrum, and trispectrum on the full sky: Scalar edition", *Phys. Rev. D* **107**, 123516 (2023) (arXiv).

34. Creque-Sarbinowski, C., Alexander, S., Kamionkowski, M., **Philcox, O. H. E.**, "Parity-violating trispectrum from Chern-Simons gravity", *JCAP* 11 029 (2023) (arXiv).

35. Surrao, K. M., **Philcox, O. H. E.**, Hill, J. C., "Accurate estimation of angular power spectra for maps with correlated masks", *Phys. Rev. D* **107**, 083521 (2023) (arXiv).

36. Ivanov, M. M., **Philcox, O. H. E.**, Cabass, G., Nishimichi, T., Simonović, M., Zaldarriaga, M., "Cosmology with the galaxy bispectrum multipoles: Optimal estimation and application to BOSS data", *Phys. Rev. D* **107**, 083515 (2023) (arXiv).

37. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, Simonovic, M., Zaldarriaga, M., "Constraining single-field inflation with MegaMapper", *Phys. Lett. B* **841**, 137912 (2023) (arXiv).

38. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, "Colliders and ghosts: Constraining inflation with the parity-odd galaxy four-point function", *Phys. Rev. D* **107**, 023523 (2023) (arXiv).

39. Goldstein, S., Esposito, A., **Philcox, O. H. E.**, Hui, L., Hill, J. C., Scoccimarro, R., Abitbol, M. H., "Squeezing $f_{\rm NL}$ out of the matter bispectrum with consistency relations", *Phys. Rev. D* **106**, 123525 (2022) (arXiv).

40. ***Philcox, O. H. E.**, Torquato, S., "Disordered Heterogeneous Universe: Galaxy Distribution and Clustering across Length Scales", *Phys. Rev. X* **13**, 011038 (2023) (arXiv).

41. **Philcox, O. H. E.**, Johnson, M. C., "Novel cosmological tests from combining galaxy lensing and the polarized Sunyaev-Zel'dovich effect", *Phys. Rev. D* **106**, 083501 (2022) (arXiv).

42. **Philcox, O. H. E.**, "Probing parity violation with the four-point correlation function of BOSS galaxies", *Phys. Rev. D* **106**, 063501 (2022) (arXiv).

43. **Philcox, O. H. E.**, Ivanov, M. M., Cabass, G., Simonović, M., Zaldarriaga, M., Nishimichi, T., "Cosmology with the redshift-space galaxy bispectrum monopole at one-loop order", *Phys. Rev. D* **106**, 043530 (2022) (arXiv).

44. **Philcox, O. H. E.**, Farren, G. S., Sherwin, B. D., Baxter, E. J., Brout, D. J., "Determining the Hubble constant without the sound horizon: A 3.6% constraint on H_0 from galaxy surveys, CMB lensing, and supernovae", *Phys. Rev. D* **106**, 063530 (2022) (arXiv).

45. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., Zaldarriaga, M., "Constraints on multifield inflation from the BOSS galaxy survey", *Phys. Rev. D* **106**, 043506 (2022) (arXiv).

46. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., Zaldarriaga, M., "Constraints on Single-Field Inflation from the BOSS Galaxy Survey", *Phys. Rev. Lett.* **129**, 021301 (2022) (arXiv).

47. *Farren, G. S., **Philcox, O. H. E.**, Sherwin, B. D., "Determining the Hubble constant without the sound horizon: Perspectives with future galaxy surveys", *Phys. Rev. D* **105**, 063503 (2022) (arXiv).

48. **Philcox, O. H. E.**, Ivanov, M. M., "BOSS DR12 full-shape cosmology: ΛCDM constraints from the large-scale galaxy power spectrum and bispectrum monopole", *Phys. Rev. D* **105**, 043517 (2022) (arXiv).

49. Ivanov, M. M., **Philcox, O. H. E.**, Nishimichi, T., Simonović, M., Takada, M., Zaldarriaga, M., "Precision analysis of the redshift-space galaxy bispectrum", *Phys. Rev. D* **105**, 063512 (2022) (arXiv).

50. Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., Zaldarriaga, M., Nischimichi, T., Takada, M., "Cosmological constraints without nonlinear redshift-space distortions", *Phys. Rev. D* **105**, 043531 (2022) (arXiv).

51. **Philcox, O. H. E.**, Hou, J., Slepian, Z., "A First Detection of the Connected 4-Point Correlation Function of Galaxies Using the BOSS CMASS Sample", *submitted to Phys. Rev. D* (arXiv).

52. **Philcox, O. H. E.**, "Cosmology without window functions. II. Cubic estimators for the galaxy bispectrum", *Phys. Rev. D* **104**, 123529 (2021) (arXiv).

53. ***Philcox, O. H. E.**, Slepian, Z., "Efficient computation of N-point correlation functions in D dimensions", *PNAS* **119**, 33 (2022) (arXiv).

54. **Philcox, O. H. E.**, Slepian, Z., Hou, J., Warner, C., Cahn, R. N., Eisenstein, D. J., "encore: an O (Ng2) estimator for galaxy N-point correlation functions", *MNRAS* **509**, 2457 – 2481 (2021) (arXiv).

55. ***Philcox, O. H. E.**, Slepian, Z., "An exact integral-to-sum relation for products of Bessel functions", *Proc. Roy. Soc. A* **477**, 2253 (2021) (arXiv).

56. Slepian, Z., **Philcox, O. H. E.**, "A uniform spherical goat (problem): explicit solution for homologous collapse's radial evolution in time", *MNRAS* **522**, L42-L45 (2023) (arXiv).

57. ***Philcox, O. H. E.**, Slepian, Z., "Beyond the Yamamoto approximation: Anisotropic power spectra and correlation functions with pairwise lines of sight", *Phys. Rev. D* **103**, 123509 (2021) (arXiv).

58. **Philcox, O. H. E.**, "Cosmology without window functions: Quadratic estimators for the galaxy power spectrum", *Phys. Rev. D* **103**, 103504 (2021) (arXiv).

59. **Philcox, O. H. E.**, Aviles, A., Massara, E., "Modeling the Marked Spectrum of Matter and Biased Tracers in Real- and Redshift-Space", *JCAP* 03 038 (2021) (arXiv).

60. **Philcox, O. H. E.**, Ivanov, M. M., Zaldarriaga, M., Simonovic, M., Schmittfull, M., "Fewer Mocks and Less Noise: Reducing the Dimensionality of Cosmological Observables with Subspace Projections", *Phys. Rev. D* **103**, 043508 (2021) (arXiv).

61. **Philcox, O. H. E.**, Sherwin, B. D., Farren, G. S., Baxter, E. J., "Determining the Hubble Constant without the Sound Horizon: Measurements from Galaxy Surveys", *Phys. Rev. D* **103**, 023538 (2021) (arXiv).

62. **Philcox, O. H. E.**, Massara, E., Spergel, D. N., "What does the marked power spectrum measure? Insights from perturbation theory", *Phys. Rev. D* **102**, 043516 (2020) (arXiv).

63. **Philcox, O. H. E.**, "A faster Fourier transform? Computing small-scale power spectra and bispectra for cosmological simulations in $\mathcal{O}(N^2)$ time", *MNRAS* **501**, 4004 – 4034 (2021) (arXiv).

64. ***Philcox, O. H. E.**, Spergel, D. N., Villaescusa-Navarro, F., "Effective halo model: Creating a physical and accurate model of the matter power spectrum and cluster counts", *Phys. Rev. D* **101**, 123520 (2020) (arXiv).

65. **Philcox, O. H. E.**, Ivanov, M. M., Simonović, M., Zaldarriaga, M., "Combining Full-Shape and BAO Analyses of Galaxy Power Spectra: A 1.6% CMB-independent constraint on H_0", *JCAP* 05 032 (2020) (arXiv).

66. **Philcox, O. H. E.**, Eisenstein, D. J., "Computing the Small-Scale Galaxy Power Spectrum and Bispectrum in Configuration-Space", *MNRAS* **492**, 1214 – 1242 (2020) (arXiv).

67. **Philcox, O. H. E.**, Eisenstein, D. J., "Estimating Covariance Matrices for Two- and Three-Point Correlation Function Moments in Arbitrary Survey Geometries", *MNRAS* **490**, 5931 – 5951 (2019) (arXiv).

68. ***Philcox, O. H. E.**, Rybizki, J., "Inferring Galactic Parameters from Chemical Abundances: A Multi-Star Approach", *ApJ* **887**, 9 (2019) (arXiv).

69. **Philcox, O. H. E.**, Eisenstein, D. J., O'Connell, R., Wiegand, A., "rascalc: a jackknife approach to estimating single- and multitracer galaxy covariance matrices", *MNRAS* **491**, 3290 – 3317 (2020) (arXiv).

70. ***Philcox, O. H. E.**, Sherwin, B. D., van Engelen, A., "Detection and Removal of B-mode Dust Foregrounds with Signatures of Statistical Anisotropy", *MNRAS* **479**, 5577 – 5595 (2018) (arXiv).

71. **Philcox, O. H. E.**, Rybizki, J., Gutcke, T., "On the Optimal Choice of Nucleosynthetic Yields, Initial Mass Function, and Number of SNe Ia for Chemical Evolution Modeling", *ApJ* **861**, 40 (2018) (arXiv).

Contributing Author

72. *Besuner, R., *et al.* (inc. **Philcox, O. H. E.**), "The Spectroscopic Stage-5 Experiment", *White Paper* (arXiv).

73. *Krause, E., *et al.* (inc. **Philcox, O. H. E.**), "A Parameter-Masked Mock Data Challenge for Beyond-Two-Point Galaxy Clustering Statistics", *ApJ* **990**, 99 (2025) (arXiv).

74. Rogers, K. K., Hložek, R., Laguë, A., Ivanov, M. M., **Philcox, O. H. E.**, *et al.*, "Ultra-light axions and the S_8 tension: joint constraints from the cosmic microwave background and galaxy clustering", *JCAP* 06 023 (2023) (arXiv).

75. *Abdalla, E., *et al.* (inc. **Philcox, O. H. E.**), "Cosmology intertwined: A review of the particle physics, astrophysics, and cosmology associated with the cosmological tensions and anomalies", *JHEA* **34**, 49 – 211 (2022) (arXiv).

76. Villaescusa-Navarro, F., *et al.* (inc. **Philcox, O. H. E.**), "The CAMELS Project: Public Data Release", *ApJS* **265**, 54 (2023) (arXiv).

77. Hou, J., Cahn, R. N., **Philcox, O. H. E.**, Slepian, Z., "Analytic Gaussian covariance matrices for galaxy N-point correlation functions", *Phys. Rev. D* **106**, 043515 (2022) (arXiv).

78. Schmittfull, M., Simonović, M., Ivanov, M. M., **Philcox, O. H. E.**, Zaldarriaga, M., "Modeling Galaxies in Redshift Space at the Field Level", *JCAP* 05 059 (2021) (arXiv).

79. Villaescusa-Navarro, F., *et al.* (inc. **Philcox, O. H. E.**), "The CAMELS project: Cosmology and Astrophysics with MachinE Learning Simulations", *ApJ* **915**, 71 (2021) (arXiv).

80. Wang, Y., Zhao, G.-B., Zhao, C., **Philcox, O. H. E.**, *et al.*, "The clustering of the SDSS-IV extended Baryon Oscillation Spectroscopic Survey DR16 luminous red galaxy and emission line galaxy samples: cosmic distance and structure growth measurements using multiple tracers in configuration space", *MNRAS* **498**, 3470 – 3483 (2020) (arXiv).

81. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., "Nonlinear perturbation theory extension of the Boltzmann code CLASS", *Phys. Rev. D* **102**, 063533 (2020) (arXiv).

Other Works

82. **Philcox, O. H. E.**, "An Introduction to the EFTofLSS", *Lecture Notes*.

SELECTED RESEARCH TALKS

* = *Virtual Talk*

2026 New Synergies in Multiprobe Cosmology, KITP, Santa Barbara, *Conference*

2025 41st Institut d'Astrophysique de Paris Symposium, *Conference* (Invited Talk)
Galileo Galilei Institute for Theoretical Physics, *Symposium* (Invited Plenary)
21st Rencontres de Vietnam, *Conference* (Invited Talk)
Understanding Cosmological Observations, Benasque, *Conference* (Invited Talk)
GR24 & Amaldi16, *Conference* (Invited Plenary)
Columbia University, *Physics Colloquium*
Cosmological Probes of New Physics, Notre Dame, *Conference* (Invited Discussion)
Brown University, *Theoretical Physics Seminar*
Princeton University, *Cosmology Seminar*
Particles vs. New Probes, Flatiron Institute, *Conference* (Invited Plenary)
Boston University, *High-Energy Physics Seminar*

2024 Essential Cosmology for the Next Generation IX, Mexico, *Winter School* (Invited Lecture Series)
21st Century Cosmology, Ashoka University, *Conference* (Invited Plenary)
Cosmology in the Adriatic, Split, *Conference*
University of Edinburgh, *Conference*
Lawrence Berkeley National Laboratory, *Workshop* (Invited Plenary)
University of Cambridge, *Astronomy Colloquium*
Royal Society, London, *Discussion Meeting* (Invited Talk)
Harvard University, *Cosmology Seminar*
Massachusetts Institute of Techology, *Cosmology Seminar*
58th Rencontres de Moriond, *Conference*
Stanford University, *Physics Colloquium*
Perimeter Institute for Theoretical Physics, *Colloquium*

2023 Taipei Institute of Astronomy & Astrophysics, *Conference* (Invited Talk)
Max-Planck-Institute for Nuclear Physics, Heidelberg, *Particle Physics Seminar*
ICTP Trieste, *Workshop*
University of Montreal, *Astrophysics Seminar*
*Parity-Violation from Home, *Conference*
Ohio State University, *Cosmology and Astro-Particle Physics Seminar*
*Early Universe / AliCPT Forum, University of Science and Technology of China, *Webinar*
Sexten Center for Astrophysics, *Workshop* (Invited Plenary)
Donostia International Physics Center, *Workshop* (Invited Talk)
University of Pennsylvania, *PDT Partners Retreat* (Invited Talk)
Yukawa Institute for Theoretical Physics, Kyoto, *Conference* (Invited Talk)
Kavli IPMU, Tokyo, *Astronomy Seminar*
Stony Brook, *Cosmology Seminar*
Cosmology on Safari, *Conference*
Johns Hopkins University, *Particle Physics Seminar*
University of Maryland, *Particle Physics Seminar*
*Copernicus Series, *Cosmology Webinar*
*University of Oxford, *Cosmology Seminar*

OUTREACH TALKS

2026	*Astronomy on Tap*, San Francisco (Public Talk) *Discover Our Universe*, Kavli Institute for Particle Astrophysics & Cosmology (Public Lecture)
2025	*Stanford University Undergraduate Physics Club* (Faculty Lunch Talk) *Taste of Science*, New York (Public Lecture)
2023	*Into The Unknown* (Public Podcast)

MEDIA

1. "Scientists detect mysterious suppression in cosmic structure growth", *Phys.org*, 3 Jan. 2025.
2. "Universe's 'Cosmological Collider' Lands 3 Scientists $100,000 Physics Prize", *Live Science*, 14 Sep. 2023.
3. "What Happened Right After the Universe Began?", *Simons Foundation*, 30 Aug. 2023.
4. "The Cosmos as a Colloid", *Physics Magazine*, 14 Mar. 2023.
5. "Pinpoint Simulations Provide Perspective on Universe Structure", *IAS News* & *Phys.org*, 14 Mar. 2023.
6. "Spatial Pattens In Distribution of Galaxies", *Princeton News* & *ScienceDaily*, 14 Mar. 2023.
7. "Is the Universe Asymmetrical?", *Columbia News*, 27 Feb. 2023.
8. "Do We Live in a Mirror Universe?", *Into The Unknown Podcast*, 26 Jan. 2023.
9. "Asymmetry Detected in the Distribution of Galaxies", *Quanta*, 5 Dec. 2022.
10. "The Universe is Surprisingly Lopsided and We Don't Know Why", *New Scientist*, 18 Jun. 2022.

PRIZES & GRANTS

2024	New Horizons in Physics Prize, *Breakthrough Prize Foundation*
2023	Buchalter Cosmology Prize (First Prize)
2022	Simons Society of Fellows (Junior Fellowship, $450 000) NHFP Einstein Fellowship, *declined* LBL Chamberlain Fellowship, *declined* Cambridge Kavli Fellowship, *declined*
2018	Herchel-Smith Scholarship ($70 000), *Cambridge* → *Harvard* Institute of Astronomy Prize, *Cambridge*
2017	Holgate Pollard Memorial Prize, *Cambridge*

gi **PROFESSIONAL ACTIVITIES**

Referee	MNRAS (2020−), JCAP (2020−), MPLA (2021−), Phys. Rev. Lett. (2022−) Phys. Rev. D (2022−), ApJS (2022−), Phys. Dark Univ. (2023−)	
Conferences	Large-Scale Parity Violation (Taiwan 2023, SOC)	
Review Panels	NASA Astrophysics Theory Program (2023)	
Mentor	*Beatriz Tucci*	Stanford Postdoctoral Fellow (2025−)
Advisor	*Cooper Jacobus*	Stanford Graduate Student (2026, rotation)
	Hanyue Wang	Stanford Graduate Student (2026, rotation)
	Sixiang Yang	Stanford Graduate Student (2025, rotation)
	Alessandro Russo	Stanford Graduate Student (2025, rotation)
	Anita Dunsmore	Stanford Graduate Student (2025, rotation)
	Sam Goldstein	Columbia Graduate Student (2022−2025, various projects)
	John Moynihan	Columbia Undergraduate Student (2023, summer project)
	Jess Boyland	Simons-NSBP Undergraduate Scholar (2020−2021, summer project)
Coadvisor	*Luca Abu El-Haj*	Columbia Undergraduate Student (2025, summer project)
	Gemma Zhang	Harvard Graduate Student (2023−2025)
	Kristen Surrao	Columbia Graduate Student (2022−2023)
	Gerrit Farren	Cambridge Graduate Student (2020−2022)

MISCELLANEOUS

Computing Languages	Python, C++, Julia, Mathematica, Cuda
Codes Developed	PolySpec, PolyBin3D, Encore, NPCFs.jl, Class-PT, Spectra-Without-Windows, EffectiveHalos, Hipster, RascalC
Teaching	6 years of online tutoring (high-school to Masters level) Teaching assistant for Princeton introductory astronomy class (AST203) TEFL qualification in English teaching
Other	DipABRSM in Music Performance (Distinction)

REFERENCES

Prof. J C Hill
(Postdoctoral Mentor)
Columbia University
914 Pupin Hall
New York, NY 10027
Tel: +1 (212) 854-7815
jch2200@columbia.edu

Prof. D N Spergel
(PhD Advisor)
Simons Foundation
160 5th Ave.
New York, NY 10010
Tel: +1 (609) 258-3589
dspergel@simonsfoundation.org

Prof. M Zaldarriaga
(PhD Advisor)
Institute for Advanced Study
1 Einstein Drive
Princeton, NJ 08540
Tel: +1 (609) 734-8058
matiasz@ias.edu